Homework #2 Solution

1- (a) The phase condition 2nL = m λ is required for oscillation

Therefore the lasing wavelength is either

$$\frac{2\times 3.4\times(350\times 10^{-6})}{1535}\approx 1550.49\text{ nm}$$

or

$$\frac{2\times 3.4\times(350\times 10^{-6})}{1536}\approx 1549.48\text{ nm}$$

(b) the spacing is

$$\Delta\lambda=\frac{\lambda^2}{2nL}=\frac{(1550\times 10^{-9})^2}{2\times 3.4\times(350\times 10^{-6})}\approx 1.01\text{ nm}$$

2- (a) The maximum data rate is

$$R_b\le\frac{1}{4\Delta T}=\frac{2cn_1}{L(n_1-n_2)^2}=\frac{2\times 2.998\times 10^8\times 1.5}{40\times 10^3\times(1.5-1.485)^2}\approx 100\text{ Mb/s}$$

(b) the spacing is

$$R_b\le\frac{1}{4\Delta T}=\frac{2cn_1}{L(n_1-n_2)^2}=\frac{2\times 2.998\times 10^8\times 1.5}{400\times 10^3\times(1.49-1.48)^2}\approx 22.4\text{ Mb/s}$$

3- (a) 100-km propagation: 0.22×100+1 = 23 dBm

200-km propagation: 0.22×200+1 = 45 dBm

300-km propagation: 0.22×300+1 = 67 dBm

400-km propagation: 0.22×400+1 = 89 dBm

(b) 0.22×200+9$\times 0.25+2\times 0.5+1=48.25$ dBm

4- (a)

$\Delta T=D\cdot L\cdot\Delta\lambda=17\times 500\times 0.1=850$ ps

$$R_b\le\frac{1}{4\Delta T}=\frac{1}{4\times 850\times 10^{-12}}\approx 0.294\text{ Gb/s}$$

(b)

$$\Delta T=\frac{1}{4R_b}=\frac{1}{4\times 40\times 10^9}\approx 6.25\text{ ps}$$

$$L=\frac{6.25}{17\times 0.08}\approx 4.596\text{ km}$$

5-

$$17\times(1560-1550)\times 100=17\text{ ns}$$